Statistical Foundations of Machine Learning

TA Session 10 - Slow Paced

6.30 – 7.30 pm, 31st Oct 2026

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1 Maximum a Posteriori (MAP) estimation

Problem 1

Problem:
Let $X \sim \text{Binomial}(n, \theta)$ represent the number of successes observed in n independent Bernoulli trials with unknown parameter $\theta \in [0, 1]$. Suppose θ has a Beta prior distribution with hyperparameters $\alpha > 1$ and $\beta > 1$, whose probability density function is:

$$f(\theta) = \frac{1}{\mathrm{B}(\alpha, \beta)} \theta^{\alpha-1}(1-\theta)^{\beta-1}, \quad 0 \leq \theta \leq 1$$

where $\mathrm{B}(\alpha, \beta)$ is the Beta function. Find the Maximum A Posteriori (MAP) estimator of θ.

Solution:
The likelihood of observing $X = k$ successes is given by:

$$P(X = k \mid \theta) = \binom{n}{k} \theta^k (1-\theta)^{n-k}$$

By Bayes' theorem, the posterior probability density of θ given $X = k$ is proportional to the product of the likelihood and the prior:

$$p(\theta \mid X = k) \propto P(X = k \mid \theta) f(\theta) \propto \theta^k (1-\theta)^{n-k} \cdot \theta^{\alpha-1}(1-\theta)^{\beta-1} = \theta^{k+\alpha-1}(1-\theta)^{n-k+\beta-1}$$

Taking the natural logarithm of the unnormalized posterior:

$$\ln p(\theta \mid X = k) = (k + \alpha - 1) \ln \theta + (n - k + \beta - 1) \ln(1-\theta) + C$$

where C is a constant independent of θ.

Differentiating with respect to θ and setting the derivative equal to zero:

$$\frac{d}{d\theta} \ln p(\theta \mid X = k) = \frac{k + \alpha - 1}{\theta} - \frac{n - k + \beta - 1}{1 - \theta} = 0$$

Cross-multiplying:

$$(k + \alpha - 1)(1 - \theta) = (n - k + \beta - 1)\theta$$
$$(k + \alpha - 1) - (k + \alpha - 1)\theta = (n - k + \beta - 1)\theta$$
$$k + \alpha - 1 = (n + \alpha + \beta - 2)\theta$$

Solving for θ:

$$\hat{\theta}_{\text{MAP}} = \frac{\mathbf{k} + \alpha - \mathbf{1}}{\mathbf{n} + \alpha + \beta - \mathbf{2}}$$

Because $\alpha > 1$ and $\beta > 1$, the powers are strictly positive, ensuring this stationary point lies inside $(0, 1)$ and corresponds to the unique global maximum.

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Problem 2

Problem:
Let $X_1, X_2, \ldots, X_n$ be an independent and identically distributed random sample of component operational lifetimes, where each X_i follows an Exponential distribution with unknown rate $\lambda > 0$:

$$f(x \mid \lambda) = \lambda e^{-\lambda x}, \quad x \geq 0$$

Assume an Exponential prior distribution on λ with known hyperparameter $\beta > 0$:

$$f(\lambda) = \beta e^{-\beta \lambda}, \quad \lambda > 0$$

Derive the Maximum A Posteriori (MAP) estimator $\hat{\lambda}_{\text{MAP}}$.

Solution:

The joint likelihood of the n independent observations is:

$$L(\lambda) = \prod_{i=1}^{n} \lambda e^{-\lambda X_i} = \lambda^n \exp\left(-\lambda \sum_{i=1}^{n} X_i\right)$$

Multiplying the joint likelihood by the prior $f(\lambda) = \beta e^{-\beta \lambda}$:

$$p(\lambda \mid X_1, \ldots, X_n) \propto L(\lambda) f(\lambda) \propto \lambda^n \exp\left(-\lambda \sum_{i=1}^{n} X_i - \beta \lambda\right) = \lambda^n \exp\left(-\lambda \left(\sum_{i=1}^{n} X_i + \beta\right)\right)$$

Taking the natural logarithm:

$$g(\lambda) = n \ln \lambda - \lambda \left(\sum_{i=1}^{n} X_i + \beta\right)$$

Setting the first derivative with respect to λ to zero:

$$\frac{d}{d\lambda} g(\lambda) = \frac{n}{\lambda} - \left(\sum_{i=1}^{n} X_i + \beta\right) = 0$$

Solving for λ:

$$\hat{\lambda}_{\text{MAP}} = \frac{\mathbf{n}}{\sum_{\mathbf{i=1}}^{\mathbf{n}} \mathbf{X_i} + \beta}$$

Since $\dfrac{d^2}{d\lambda^2} g(\lambda) = -\dfrac{n}{\lambda^2} < 0$, this point is the unique maximum.

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2 Hypothesis Testing

Problem 1

Problem:
A semiconductor company tests whether a newly designed fabrication process reduces the defect rate of chips. The baseline defect rate under the current process is $p = 0.10$.

(a) Formulate the null hypothesis H_0 and alternative hypothesis H_1

(b) In the context of this test, describe in words what constitutes a *Type I error* and a *Type II error*.

(c) A quality engineer proposes a decision rule where H_0 is *never* rejected regardless of the experimental sample data. Determine the values of α and β

Solution:

(a) Hypothesis Formulation

- Null Hypothesis (H_0): $p = 0.10$ (the default status quo: no improvement in defect rate)
- Alternative Hypothesis (H_1): $p < 0.10$ (the claim of improvement).

(b) Error Definitions:

- **Type I Error (α):** Rejecting H_0 when H_0 is true. Here, it means concluding that the new fabrication process reduces defects when in reality its defect rate is still 0.10. (Risk: incurring unnecessary factory retooling costs for no actual gain).
- **Type II Error (β):** Failing to reject H_0 when H_0 is false. Here, it means failing to detect that the new process is genuinely superior, mistakenly maintaining the old process. (Risk: missing out on a valuable engineering improvement).

(c) Error Values and Trade-off:

- If H_0 is never rejected, the probability of falsely rejecting H_0 is strictly $\boldsymbol{\alpha = 0}$.
- However, if H_0 is false (e.g., true $p = 0.05$), the probability of failing to reject H_0 is $\boldsymbol{\beta = 1.0}$.

Problem 2

Problem:
A web platform tests whether an updated user interface increases sign-up rates beyond the existing rate of $p_0 = 0.20$. A random cohort of $n = 10$ visitors is observed. Let X denote the number of visitors who register. We test:

$$H_0 : p = 0.20 \quad \text{versus} \quad H_1 : p > 0.20$$

(a) Suppose the rejection region is defined as $X \geq 4$. Calculate the exact significance level α.

(b) For this rejection region, compute the Type II error probability $\beta(0.50)$ and the power of the test if the true sign-up rate is $p = 0.50$.

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(c) If an experiment yields $X = 5$ sign-ups, compute the exact p-value and determine the test decision at $\alpha = 0.05$.

Solution:

(a) Significance Level α:
Under H_0, $X \sim \text{Binomial}(n = 10, p = 0.20)$. The significance level is the probability of falling into the rejection region under H_0:

$$\alpha = P(X \geq 4 \mid p = 0.20) = 1 - \sum_{k=0}^{3} \binom{10}{k} (0.2)^k (0.8)^{10-k}$$

Evaluating the individual binomial terms:

$$P(X = 0) = (0.8)^{10} \approx 0.1074$$
$$P(X = 1) = 10(0.2)(0.8)^9 \approx 0.2684$$
$$P(X = 2) = 45(0.2)^2(0.8)^8 \approx 0.3020$$
$$P(X = 3) = 120(0.2)^3(0.8)^7 \approx 0.2013$$

Summing the lower terms:

$$P(X \leq 3) = 0.1074 + 0.2684 + 0.3020 + 0.2013 = 0.8791$$
$$\alpha = \mathbf{1 - 0.8791 = 0.1209} \quad (\mathbf{12.09}\%)$$

(b) Type II Error $\beta(0.50)$ and Power:
Under the alternative $p = 0.50$, $X \sim \text{Binomial}(10, 0.50)$. A Type II error occurs when X falls in the non-rejection region ($X \leq 3$):

$$\beta(0.50) = P(X \leq 3 \mid p = 0.50) = \sum_{k=0}^{3} \binom{10}{k} (0.5)^{10} = \frac{1 + 10 + 45 + 120}{1024} = \frac{176}{1024} \approx \mathbf{0.1719}$$

The power of the test at $p = 0.50$ is:

$$\text{Power} = 1 - \beta(0.50) = 1 - 0.1719 = \mathbf{0.8281} \quad (\mathbf{82.81}\%)$$

(c) p-Value for Observed $X = 5$:
The p-value is the probability, computed assuming H_0 is true, of observing a test statistic at least as extreme in the direction of H_1 as $x = 5$:

$$p\text{-value} = P(X \geq 5 \mid p = 0.20) = 1 - P(X \leq 4 \mid p = 0.20)$$

Adding the term for $k = 4$:

$$P(X = 4) = \binom{10}{4} (0.2)^4 (0.8)^6 = 210(0.0016)(0.2621) \approx 0.0881$$
$$P(X \leq 4) = 0.8791 + 0.0881 = 0.9672$$
$$\mathbf{p}\text{-value} = \mathbf{1 - 0.9672 = 0.0328}$$

Decision: Since p-value $= 0.0328 \leq \alpha = 0.05$, we **reject** H_0 at the 5% significance level.

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